N22 Laplace eigenvalue problem

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We solve Eq.(2.1) in Filippo Bigi's part II thesis by finding out the ladder operators here. We wish to solve the Laplace eigenvalue problem

$$-\nabla^2\psi_{\kappa\ell m}(r,\theta,\phi) = \kappa^2\psi_{n\ell m}(r,\theta,\phi) \tag{1.1}$$

Consider the time-independent Schrödinger equation of a free particle in 3D

$$H|\kappa,\ell,m\rangle = E_\kappa|\kappa,\ell,m\rangle = \frac{\hbar^2\kappa^2}{2\mu}|\kappa,\ell,m\rangle \tag{1.2}$$

where H is simply radial kinetic energy plus angular kinetic energy (No potential, since it is free)

$$H = \frac{p_r^2}{2\mu} + \frac{L^2}{2\mu r^2} \tag{1.3}$$

where μ is the mass of the particle, and r is the distance to origin, the radial momentum is defined as

$$p_r = \frac{1}{2}\left(\hat{\mathbf{r}}\cdot\mathbf{p} + \mathbf{p}\cdot\hat{\mathbf{r}}\right) \tag{1.4}$$

where $\hat{\mathbf{r}}$ is a unit vector radial position operator, and $\mathbf{p}$ is the momentum operator in 3D. The canonical commutation relation $[x,p] = i\hbar$ analogue for radial direction is

$$[r,p_r] = i\hbar \tag{1.5}$$

now recognise

$$\psi_{\kappa\ell m}(r,\theta,\phi) = \langle r,\theta,\phi|\kappa,\ell,m\rangle \tag{1.6}$$

and recognise the factorise the ket into radial and angular part, just like a hydrgoen atom

$$|\kappa,\ell,m\rangle = |\kappa,\ell\rangle|\ell,m\rangle \tag{1.7}$$

and also

$$|r,\theta,\phi\rangle = |r\rangle|\theta,\phi\rangle \tag{1.8}$$

where $|n,\ell\rangle$ is the eigenket of the radial Hamiltonian H_ℓ

$$H_\ell \equiv \frac{p_r^2}{2\mu} + \frac{\ell(\ell+1)\hbar^2}{2\mu r^2} \tag{1.9}$$

so the reason that ℓ apears in the radial ket is because ℓ controls the effective potential in H_ℓ. $|\ell,m\rangle$ is a mutual eigenstate of L^2 and L_z, and the inner product

$$Y_\ell^m(\theta,\phi) = \langle\theta,\phi|\ell,m\rangle \tag{1.10}$$

is known as spherical harmonics. The reason we conisder this Hamiltonian is because

$$-\frac{\hbar^2}{2\mu}\nabla^2\langle r,\theta,\phi|\psi\rangle = \langle r,\theta,\phi|H|\psi\rangle \tag{1.11}$$

Since the angular part $|\ell,m\rangle$ is already solved by introducing ladder operators in any elementary quantum mechanics course (or see N2 Ladder operators), we are only left with solving the radial eigenvalue problem

$$H_\ell|\kappa,\ell\rangle = E_\kappa|\kappa,\ell\rangle \tag{1.12}$$

Now introduce the ladder operators

$$A_\ell = \frac{\ell+1}{r} - \frac{i}{\hbar}p_r, \qquad A_\ell^\dagger = \frac{\ell+1}{r} + \frac{i}{\hbar}p_r \tag{1.13}$$

it is easy to show that

$$A_\ell^\dagger A_\ell = \frac{2\mu}{\hbar^2}H_\ell \tag{1.14}$$

and

$$A_\ell A_\ell^\dagger = \frac{2\mu}{\hbar^2}H_{\ell+1} \tag{1.15}$$

so

$$\begin{aligned} H_{\ell+1}A_\ell|n,\ell\rangle &= \frac{\hbar^2}{2\mu}A_\ell A_\ell^\dagger A_\ell|n,\ell\rangle \\ &= A_\ell H_\ell|n,\ell\rangle \\ &= E_\kappa A_\ell|n,\ell\rangle \end{aligned} \tag{1.16}$$

thus we can conclude

$$A_\ell|\kappa,\ell\rangle \propto |\kappa,\ell+1\rangle \tag{1.17}$$

similarly by considering $H_\ell A_\ell^\dagger|n,\ell+1\rangle$

$$A_\ell^\dagger|\kappa,\ell+1\rangle \propto |\kappa,\ell\rangle \tag{1.18}$$

To obtain radial position representation of $|\kappa,\ell\rangle$ consider the case when $\ell=0$, we have

$$\frac{1}{2\mu}\langle r|p_r^2|\kappa,0\rangle = \kappa^2\langle r|\kappa,0\rangle \tag{1.19}$$

use the position representation of p_r (see N2 ladder operator, or Binney and Skinners Eq.(7.100))

$$\langle r|p_r|\psi\rangle = -i\hbar\left(\frac{\partial}{\partial r}+\frac{1}{r}\right)\langle r|\psi\rangle \tag{1.20}$$

and let

$$u(r) = r\langle r|\kappa,0\rangle \tag{1.21}$$

then it it easy to show that

$$u''(r) = -\kappa^2 u(r) \tag{1.22}$$

clearly, this is solved by $\sin\kappa r$, or $\cos\kappa r$, or a linear combination of them, thus

$$\langle r|\kappa,0\rangle = A\frac{\sin\kappa r}{r} + B\frac{\cos\kappa r}{r} \tag{1.23}$$

where A and B are coefficients. As $r \to 0$, $\langle r|\kappa, 0\rangle$ should stay finite, but the second term goes as $1/r$, so it cannot exist, and we have

$$\langle r|\kappa, 0\rangle \propto \frac{\sin \kappa r}{r} \tag{1.24}$$

Note the following sinc limit of Dirac delta

$$\delta(r) = \lim_{\kappa \to 0} \frac{1}{\pi} \frac{\sin \kappa r}{r} \tag{1.25}$$

we now normalise $|\kappa, 0\rangle$ by

$$\begin{aligned} \delta(\kappa - \kappa') &= \langle \kappa, 0|\kappa', 0\rangle \\ &= N^2 \int_0^\infty r^2 dr \, \langle \kappa, 0|r\rangle\langle r|\kappa', 0\rangle \\ &= N^2 \int_0^\infty dr \, \sin(\kappa r) \sin(\kappa' r) \end{aligned} \tag{1.26}$$

We will now do this integral in the following way (mathematician can skip over the next bit to reduce annoyance). Consider the particle in a box of size L wavefunction (Atkins' molecular quantum mechanics Eq.(2.31)), which is normalised in the following way

$$\delta_{nm} = \langle n|m\rangle = \int_0^L dx \, \langle n|x\rangle\langle x|m\rangle = \frac{2}{L} \int_0^L dx \, \sin(\kappa_n x) \sin(\kappa_m x) \tag{1.27}$$

where

$$\kappa_n = \frac{n\pi}{L} = n\Delta\kappa \tag{1.28}$$

so

$$\frac{L}{2}\delta_{nm} = \frac{\pi}{2} \frac{\delta_{nm}}{\delta\kappa} \tag{1.29}$$

then consider for any smooth function f

$$\sum_m \Delta\kappa \frac{\delta_{nm}}{\Delta\kappa} f(\kappa_m) = f(\kappa_n) \tag{1.30}$$

now take $L \to \infty$ we have

$$\int d\kappa' \, \delta(\kappa - \kappa') f(\kappa') = f(\kappa) \tag{1.31}$$

thus we can identify

$$\delta(\kappa - \kappa') = \frac{1}{\pi} \lim_{L \to \infty} (L\delta_{nm}) \tag{1.32}$$

substitute this back into Eq.(1.27) gives

$$\int_0^\infty dr \, \sin(\kappa r) \sin(\kappa' r) = \frac{\pi}{2} \delta(\kappa - \kappa') \tag{1.33}$$

since the integrand is symmetry with respect to $r \to -r$ we also have

$$\int_{-\infty}^\infty dr \, \sin(\kappa r) \sin(\kappa' r) = \pi \delta(\kappa - \kappa') \tag{1.34}$$

thus $N^2 = 2/\pi$ and the normalised wavefunction is

$$\langle r|\kappa, 0\rangle = \sqrt{\frac{2}{\pi}}\frac{\sin \kappa r}{r} \tag{1.35}$$

to normalise every other states, we do

$$\begin{aligned}\langle \kappa, \ell+1|\kappa', \ell+1\rangle &= N^2\langle \kappa, \ell|A_\ell^\dagger A_\ell|\kappa', \ell\rangle \\ &= N^2\frac{2\mu}{\hbar^2}\langle \kappa, \ell|H_\ell|\kappa', \ell\rangle \\ &= N^2\kappa^2\langle \kappa, \ell|\kappa', \ell\rangle \\ &= N^2\kappa^2\delta(\kappa - \kappa')\end{aligned} \tag{1.36}$$

thus $N = 1/\kappa$ and

$$|\kappa, \ell+1\rangle = \frac{1}{\kappa}A_\ell|\kappa, \ell\rangle \tag{1.37}$$

we can now generate normalised $|\kappa, \ell\rangle$ in the following way

$$|\kappa, \ell\rangle = \frac{1}{\kappa^\ell}A_{\ell-1}A_{\ell-2}\cdots A_0|\kappa, 0\rangle \tag{1.38}$$

now define radial wavefunction

$$R_\ell(r) = \langle r|\kappa, \ell\rangle \tag{1.39}$$

the ladder operator in r-representation

$$A_\ell = \frac{\ell}{r} - \frac{\partial}{\partial r} \tag{1.40}$$

There is one further simplification if we let $x = \kappa r$ then

$$R_{\ell+1}(x) = \left(\frac{\ell}{x} - \frac{\partial}{\partial x}\right)R_\ell(x) = -x^\ell\frac{\partial}{\partial x}\frac{R_\ell(x)}{x^\ell} \tag{1.41}$$

divide by $x^{\ell+1}$ on both sides

$$\frac{R_{\ell+1}(x)}{x^{\ell+1}} = -\frac{1}{x}\frac{\partial}{\partial x}\frac{R_\ell(x)}{x^\ell} \equiv D\left(\frac{R_\ell(x)}{x^\ell}\right) \tag{1.42}$$

where we have define an operator D, and it is worth emphasising that unlike A_ℓ, D is independent of ℓ. We now have an recurance relation, we can then repeatedly substitute the LHS into RHS, with ℓ decreasing by one each time until $\ell = 0$, we have

$$\frac{R_\ell(x)}{x^\ell} = D^\ell R_0(x) \tag{1.43}$$

moving x^ℓ to RHS, then writing it more explicitly gives

$$j_\ell(x) = (-x)^\ell\left(\frac{1}{x}\frac{\partial}{\partial x}\right)^\ell\frac{\sin x}{x} \tag{1.44}$$

the expression is known as **Rayleigh's formulas**, which in fact just a change of variable of our ladder operators in the radial representation, and $j_\ell(x)$ is known as **spherical Bessel functions**, they are related to the wavefunction by

$$R_\ell(r) = \sqrt{\frac{2}{\pi}}\kappa j_\ell(\kappa r) \tag{1.45}$$

thus we have solved eigenvalue equation of ∇^2.

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We now solve for the radial part with boundary condition of the radial function vanishes at a hard sphere boundary of radius a, $R_\ell(a) = 0$, thus $j_\ell(\kappa a) = 0$. Clearly, much like particle in a box, energy and consequently κ now becomes quantised, and we shall relabel the radial eigenstate $|\kappa, \ell\rangle \to |n, \ell\rangle$, where n is an integer, and consequently $\langle n, \ell | n', \ell\rangle = \delta_{nn'}$. For $\ell = 0$, boundary condition requires

$$\sin(\kappa a) = 0 \tag{2.1}$$

thus for $\ell = 0$

$$\kappa = \frac{n\pi}{a}, \qquad n = 0, 1, 2, \ldots \tag{2.2}$$

so we should also relabel $\kappa \to \kappa_{n\ell}$, we also define $z_{n\ell} = \kappa_{n\ell} a$ to label the roots of j_ℓ, energy becomes

$$E_{n\ell} = \frac{\hbar^2 \kappa_{n\ell}}{2\mu} = \frac{\hbar^2 z_{n\ell}^2}{2\mu a^2} \tag{2.3}$$

subtlety j_ℓ does not acquire a new label n because the label n comes from the argument of j_ℓ rather than its functional form. The radial wavefunction is

$$\langle r | n, \ell\rangle = N_{n\ell} j_\ell(\kappa_{n\ell} r) \tag{2.4}$$

We now normalise $|n, 0\rangle$ by

$$\begin{aligned} 1 &= \langle n, 0 | n, 0\rangle \\ &= \int_0^a r^2 \, dr \, \langle n, 0 | r\rangle \langle r | n, 0\rangle \\ &= N^2 \int_0^a dr \, \sin^2 \frac{n\pi r}{a} \end{aligned} \tag{2.5}$$

which is the same as integration particle in a box wavefunction, so

$$\langle r | n, 0\rangle = \sqrt{\frac{2}{a}} \frac{\sin(n\pi r/a)}{r} \tag{2.6}$$

in general (we will not derive here)

$$\langle r | n, \ell\rangle = \frac{\sqrt{2}}{a^{3/2} |j_{\ell+1}(z_{n\ell})|} j_\ell\left(\frac{z_{n\ell} r}{a}\right) \tag{2.7}$$